Macroeconomics III - Assignment 1

Emil Oliver Olesen (fps618) & Carl-Emil Rohde (kvb125)

feb 2026

1 Exercise 1

1.1 a)

We want to find the optimal consumption path meaning how much to consume each period. To do this, we set up a Lagrangian and take derivatives. Our constraint each period is: $c_t + a_{t+1} = y_t + (1+r)a_t$, which we rearrange: $y_t + (1+r)a_t - c_t - a_{t+1} = 0$ Since this constraint holds for every period, we need a separate multiplier λ_t for each one:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \underbrace{\frac{c_t^{1-\sigma}}{1-\sigma}}_{\text{utility}} + \sum_{t=0}^{\infty} \lambda_t \left[y_t + (1+r)a_t - c_t - a_{t+1} \right]$$

We have two choice variables each period: $\mathbf{c_t}$ (consumption) and $\mathbf{a_{t_{+1}}}$ (savings) so we find FOCs with respect for both.

1.1.1 FOC with respect to c_t

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t \cdot \frac{(1-\sigma)c_t^{-\sigma}}{1-\sigma} + \lambda_t \cdot (-1) = \beta^t c_t^{-\sigma} - \lambda_t$$

Setting the sum of both terms equal to zero:

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t c_t^{-\sigma} - \lambda_t = 0$$

$$\boxed{\lambda_t = \beta^t c_t^{-\sigma}}$$

1.1.2 FOC with respect to $a_{t_{+1}}$

Because $\mathbf{a_{t_{+1}}}$ appears in two different period constraints, we need to differentiate term by term:

Period t:

$$\frac{\partial}{\partial a_{t+1}} \lambda_t \left[y_t + (1+r)a_t - c_t - a_{t+1} \right] = \lambda_t \cdot (-1) = -\lambda_t$$

Period $t+1$

$$\frac{\partial}{\partial a_{t+1}}\lambda_{t+1}\left[y_{t+1}+(1+r)a_{t+1}-c_{t+1}-a_{t+2}\right]=\lambda_{t+1}\cdot(1+r)$$

Setting the sum of both terms equal to zero gives us the second FOC:

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}}=-\lambda_t+(1+r)\lambda_{t+1}=0$$

$$\boxed{\lambda_t=(1+r)\lambda_{t+1}}$$

1.1.3 Obtaining the euler-equation

To better understand the trade off between consumption and saving, we can substitute the c_t FOC into the a_t+t FOC.

From the c_t FOC we got:

$$\lambda_t=\beta^t c_t^{-\sigma}$$

This holds for any period as it's a general statement. So if we shift it forward one period we get:

$$\lambda_{t+1}=\beta^{t+1}c_{t+1}^{-\sigma}$$

These two terms for λ_t and λ_t+1 can now be inserted directly into the second FOC:

$$\underbrace{\beta^t c_t^{-\sigma}}_{\lambda_t}=(1+r)\cdot\underbrace{\beta^{t+1}c_{t+1}^{-\sigma}}_{\lambda_{t+1}}$$

Finally we divide with β^t on both sides to arrive at the euler equation:

$$\boxed{c_t^{-\sigma}=\beta(1+r)\cdot c_{t+1}^{-\sigma}}$$

$$\frac{c_{t+1}}{c_t}=[\beta(1+r)]^{1/\sigma}$$

The Euler equation tells us exactly what influences the tradeoff between consuming today vs. tomorrow adjusted for impatience β and interest $1+r$. σ is an elasticity parameter. A higher σ means you're less willing to substitute across time even if r is high, you won't shift much consumption to the future.

At the optimum, you're indifferent between consuming one unit today or saving it and consuming (1+r) units tomorrow. If this weren't true, you could make yourself better off by shifting consumption between periods → thus you have not found the optimal bundle.

1.2 b)

No-Ponzi Game Condition The no-Ponzi game (NPG) condition is

$$\lim_{T\to\infty} \frac{a_{T+1}}{(1+r)^{T+1}} \geq 0.$$

It rules out the possibility that the consumer can finance consumption by rolling over debt forever. Hence, it prevents explosive borrowing schemes. The NPG condition is a feasibility constraint as it is imposed on the agent from outside, not something they choose. It restricts the set of feasible plans the agent can choose from.

Present-Value Budget Constraint Starting from the period budget constraint

$$a_{t+1} = (1+r)a_t + y_t - c_t,$$

forward substitution yields

$$a_{T+1} = (1+r)^{T+1}a_0 + \sum_{t=0}^{T}(1+r)^{T-t}(y_t - c_t).$$

Dividing by $(1+r)^{T+1}$ gives

$$\frac{a_{T+1}}{(1+r)^{T+1}} = a_0 + \sum_{t=0}^{T}\frac{y_t - c_t}{(1+r)^{t+1}}.$$

Taking the limit as $T \to \infty$ and using the NPG condition together with the transversality condition (which implies that the discounted value of assets converges to zero), we obtain

$$0 = a_0 + \sum_{t=0}^{\infty}\frac{y_t - c_t}{(1+r)^{t+1}}.$$

Rearranging yields the intertemporal budget constraint:

$$\boxed{\sum_{t=0}^{\infty}\frac{c_t}{(1+r)^{t+1}} = a_0 + \sum_{t=0}^{\infty}\frac{y_t}{(1+r)^{t+1}}}$$

Thus, the discounted value of lifetime consumption equals the discounted value of lifetime resources (initial assets plus endowments).

1.3 c)

Solving for c_0 From the Euler equation we found

$$\frac{c_{t+1}}{c_t} = [\beta(1+r)]^{\frac{1}{\sigma}}.$$

Iterating this forward gives the entire consumption path as a function of c_0:

$$c_t = \left([\beta(1+r)]^{\frac{1}{\sigma}}\right)^t c_0 = [\beta^t(1+r)^t]^{\frac{1}{\sigma}} c_0.$$

Now insert this expression into the intertemporal budget constraint:

$$\sum_{t=0}^{\infty} \frac{c_t}{(1+r)^{t+1}} = a_0 + \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^{t+1}}.$$

Substituting for c_t gives

$$\sum_{t=0}^{\infty} \frac{[\beta^t(1+r)^t]^{\frac{1}{\sigma}} c_0}{(1+r)^{t+1}} = a_0 + \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^{t+1}}.$$

Multiply both sides by $(1+r)$:

$$\sum_{t=0}^{\infty} \frac{1}{(1+r)^t} [\beta^t(1+r)^t]^{\frac{1}{\sigma}} c_0 = (1+r)a_0 + \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t}.$$

Rearranging,

$$\boxed{\sum_{t=0}^{\infty} \frac{1}{(1+r)^t} y_t = \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} [\beta^t(1+r)^t]^{\frac{1}{\sigma}} c_0 - (1+r)a_0}$$

In the special case $a_0 = 0$, this simplifies to

$$\boxed{\lim_{T\to\infty} \sum_{i=0}^{T} \frac{1}{(1+r)^i} y_i = \lim_{T\to\infty} \sum_{i=0}^{T} \frac{1}{(1+r)^i} \left[\beta^i(1+r)^i\right]^{\frac{1}{\sigma}} c_0}$$

which is exactly what we wanted to show.

Solving explicitly for c_0 Define

$$\Gamma = \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} [\beta^t(1+r)^t]^{\frac{1}{\sigma}}.$$

Then

$$c_0 = \frac{(1+r)a_0 + \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t}}{\Gamma}.$$

Thus, initial consumption equals total lifetime resources divided by a term that depends on preferences and the interest rate.

Marginal Propensity to Consume (MPC) The MPC out of current income is

$$\frac{\partial c_0}{\partial y_0} = \frac{1}{\Gamma}.$$

Since $\Gamma > 1$ (because future consumption also enters), we obtain

$$\boxed{0 < \frac{\partial c_0}{\partial y_0} < 1}$$

Economic Interpretation The MPC is smaller than one because the household smooths consumption over time. An unexpected increase in y_0 raises lifetime wealth, but the consumer spreads this increase across all future periods according to the Euler equation.

If $\beta(1+r)$ is close to one (strong smoothing), the MPC is small. If the consumer is very impatient or σ is small (high intertemporal substitution), the MPC is larger.

1.4 d)

From part (c) we found that the marginal propensity to consume is

$$\frac{\partial c_0}{\partial y_0} = \frac{1}{\Gamma},$$

where

$$\Gamma = \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} [\beta^t (1+r)^t]^{\frac{1}{\sigma}}.$$

Step 1: Simplifying Γ First rewrite the term inside the sum:

$$[\beta^t (1+r)^t]^{\frac{1}{\sigma}} = \beta^{\frac{t}{\sigma}} (1+r)^{\frac{t}{\sigma}}.$$

Hence,

$$\Gamma = \sum_{t=0}^{\infty} \beta^{\frac{t}{\sigma}} (1+r)^{t\left(\frac{1}{\sigma}-1\right)}.$$

This is a geometric series with common ratio

$$q = \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}.$$

Thus,

$$\Gamma = \frac{1}{1-q}.$$

Therefore the MPC becomes

$$\boxed{\frac{\partial c_0}{\partial y_0} = 1 - q = 1 - \beta^{\frac{1}{\sigma}}(1+r)^{\frac{1}{\sigma}-1}}.$$

Step 2: Plugging in parameter values Assume:

$$\sigma = 2, \qquad \beta = 0.95, \qquad r = 0.04.$$

Then

$$q = \beta^{1/2}(1+r)^{-1/2}.$$

Compute each term:

$$\beta^{1/2} = \sqrt{0.95} \approx 0.9747,$$

$$(1+r)^{-1/2} = (1.04)^{-1/2} \approx 0.9806.$$

Thus,

$$q \approx 0.9747 \times 0.9806 \approx 0.955.$$

Therefore,

$$\boxed{\text{MPC} \approx 1 - 0.955 = 0.045.}$$

Interpretation The model predicts an MPC of roughly 4–5%, which is far below the empirical estimate of about 0.3. This is because the infinite-horizon model implies strong consumption smoothing: an income shock is optimally spread over the entire lifetime, so only a small fraction is consumed immediately.

The difference relative to the data arises because the model abstracts from important real-world features. In reality, many households are liquidity constrained, have finite lifetimes, face income risk, and differ substantially from one another. Some households are effectively hand-to-mouth and therefore display much higher MPCs. These features can generate empirical MPCs that are much closer to those observed in the data.

2 Exercise 2

2.1 a)

The representative firm maximizes

$$\pi(K_t, L_t) = AK_t^{\alpha} L_t^{1-\alpha} - w_t L_t - r_t K_t.$$

FOC:

$$\frac{\partial \pi}{\partial K_t} = \alpha A K_t^{\alpha-1} L_t^{1-\alpha} - r_t = 0 \;\Rightarrow\; r_t = \alpha A K_t^{\alpha-1} L_t^{1-\alpha}$$

$$\frac{\partial \pi}{\partial L_t} = (1-\alpha) A K_t^{\alpha} L_t^{-\alpha} - w_t = 0 \;\Rightarrow\; w_t = (1-\alpha) A K_t^{\alpha} L_t^{-\alpha}.$$

Define capital per hour worked $k_t \equiv K_t/L_t$. Then

$$r_t = \alpha A \left(\frac{K_t}{L_t}\right)^{\alpha-1} = \alpha A k_t^{\alpha-1}, \qquad w_t = (1-\alpha) A \left(\frac{K_t}{L_t}\right)^{\alpha} = (1-\alpha) A k_t^{\alpha}.$$

Thus, factor prices depend only on capital per hour worked.
Total output and labor income are

$$Y_t = A K_t^{\alpha} L_t^{1-\alpha}, \qquad w_t L_t = (1-\alpha) A K_t^{\alpha} L_t^{1-\alpha}.$$

Hence, the labor income share is

$$\frac{w_t L_t}{Y_t} = \frac{(1-\alpha) A K_t^{\alpha} L_t^{1-\alpha}}{A K_t^{\alpha} L_t^{1-\alpha}} = 1-\alpha.$$

$$\boxed{\text{Labor income share} = 1-\alpha}$$

In the Cobb–Douglas case, the labor share is constant and equal to $1-\alpha$.

2.2 b)

Figure 1 plots the labor income share defined as total employee compensation divided by gross value added for Belgium and Poland over the period 1995–2021.

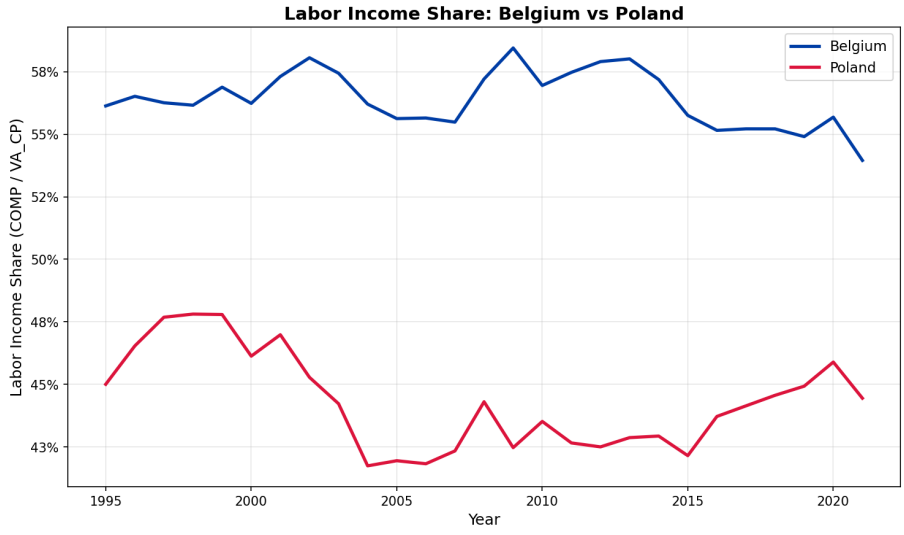


Figure 1: plots the labor income share for belgium and Poland

Belgium displays a relatively stable labor share throughout the period, averaging **56.4**%, with a modest decline toward the end of the sample. Poland exhibits a lower and more volatile labor share, averaging **44.3**%.

Table 1: Average Labor Income Share and Implied α

	Belgium	Poland
Labor income share	56.4%	44.3%
Implied α	0.436	0.557

From part (a), we showed that under Cobb-Douglas production the labor income share equals $1 - \alpha$:

$$\frac{w_t L_t}{Y_t} = 1 - \alpha$$

The average labor income share therefore directly pins down α, the capital share parameter in the production function. Using the sample averages we obtain:

$$\alpha_{BE} = 1 - 0.564 = 0.436$$
$$\alpha_{PL} = 1 - 0.443 = 0.557$$

Poland's substantially higher implied α suggests that capital accounts for a larger share of production relative to Belgium.

2.3 c)

The household chooses consumption c_t and wealth a_{t+1} to maximize utility subject to the budget constraint in every period. We attach a multiplier λ_t to each period's constraint:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t (1+n)^t \frac{c_t^{1-\sigma}}{1-\sigma} + \sum_{t=0}^{\infty} \lambda_t \left[a_t R_t + w_t - c_t - (1+n) a_{t+1} \right]$$

We take derivatives with respect to the two choice variables: c_t and a_{t+1}.

2.3.1 FOC for c_t

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t (1+n)^t c_t^{-\sigma} - \lambda_t$$
$$\beta^t (1+n)^t c_t^{-\sigma} - \lambda_t = 0$$
$$\boxed{\lambda_t = \beta^t (1+n)^t c_t^{-\sigma}}$$

2.3.2 FOC for $a_t + 1$

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}} = -\lambda_t(1+n) + \lambda_{t+1}R_{t+1}$$

$$-\lambda_t(1+n) + \lambda_{t+1}R_{t+1} = 0$$

$$\boxed{\lambda_t(1+n) = \lambda_{t+1}R_{t+1}}$$

2.3.3 Obtaining the euler-equation

To better understand the trade off between consumption and saving, we can substitute the c_t FOC into the $a_t + t$ FOC.

From the c_t FOC we got:

$$\lambda_t = \beta^t(1+n)^t c_t^{-\sigma}$$

This holds for any period as it's a general statement. So if we shift it forward one period we get:

$$\lambda_{t+1} = \beta^{t+1}(1+n)^{t+1}c_{t+1}^{-\sigma}$$

These two terms for λ_t and $\lambda_t + 1$ can now be inserted directly into the second FOC:

$$\beta^t(1+n)^t c_t^{-\sigma} \cdot (1+n) = \beta^{t+1}(1+n)^{t+1}c_{t+1}^{-\sigma} \cdot R_{t+1}$$

Cancel $(1+n)^{t+1}$ from both sides, and β^t to get the euler-equation:

$$\boxed{c_t^{-\sigma} = \beta \cdot c_{t+1}^{-\sigma} \cdot R_{t+1}}$$

The Euler equation governs when the household chooses to consume. It's the fundamental consumption-savings tradeoff.

Situation	What happens
$\beta R_{t+1} > 1$	Future is attractive $\rightarrow$ consumption grows
$\beta R_{t+1} < 1$	Present is attractive $\rightarrow$ consumption shrinks
$\beta R_{t+1} = 1$	Indifferent $\rightarrow$ consumption is flat

2.4 d)

We need to impose that markets clear meaning supply equals demand in every market. There are two markets.

Capital market: Households own the capital and rent it to firms. In per-capita terms:

$$a_t = k_t$$

Household wealth = capital stock.

Labor market: Each household supplies 1 unit of time, so aggregate labor per capita = 1:

$$L_t = 1$$

From part a of the exercise we know:

$$r_t = \alpha A k_t^{\alpha-1}$$

meaning gross returns:

$$R_t = 1 + r_t - \delta = 1 + \alpha A k_t^{\alpha-1} - \delta$$

in steady state the economy stops changing. consumption and capital per capita are constant so the growth rate of consumption;

$$\frac{c_{t+1}}{c_t} = 1$$

from part c the euler-equation is:

$$c_t^{-\sigma} = \beta \cdot c_{t+1}^{-\sigma} \cdot R_{t+1}$$

In steady state $c_{t+1} = c_t$, so $c^{-\sigma}$ cancels from both sides:

$$1 = \beta \cdot R^*$$

$$R^* = \frac{1}{\beta}$$

now we substitute in the expression for gross returns:

$$1 + \alpha A \left(k^*\right)^{\alpha-1} - \delta = \frac{1}{\beta}$$

solve for capital steadt state:

$$\alpha A \left(k^*\right)^{\alpha-1} = \frac{1}{\beta} - 1 + \delta$$

$$\boxed{k^* = \left(\frac{\alpha A}{\frac{1}{\beta} - 1 + \delta}\right)^{\frac{1}{1-\alpha}}}$$

Captial output ratio Output per capita with $L = 1$ is:

$$y^* = A \left(k^*\right)^{\alpha}$$

so the capital output ratio is:

$$\frac{k^*}{y^*} = \frac{k^*}{A\,(k^*)^{\alpha}} = \frac{(k^*)^{1-\alpha}}{A}$$

From the steady state condition above we know:

$$(k^*)^{1-\alpha} = \frac{\alpha A}{\frac{1}{\beta} - 1 + \delta}$$

Substituting:

$$\boxed{\frac{k^*}{y^*} = \frac{\alpha}{\frac{1}{\beta} - 1 + \delta}}$$

we see the capital-output ratio depends only on α, β, and δ. depreciations sits in the denominator, meaning:

$$\delta \uparrow \Rightarrow \frac{1}{\beta} - 1 + \delta \uparrow \Rightarrow \frac{k^*}{y^*} \downarrow$$

Higher depreciation makes capital more expensive to maintain. Each unit of capital loses more of its value each period, which reduces the net return to saving. Households therefore find it less attractive to accumulate capital, and the economy settles at a lower capital-output ratio in steady state.

2.5 e)

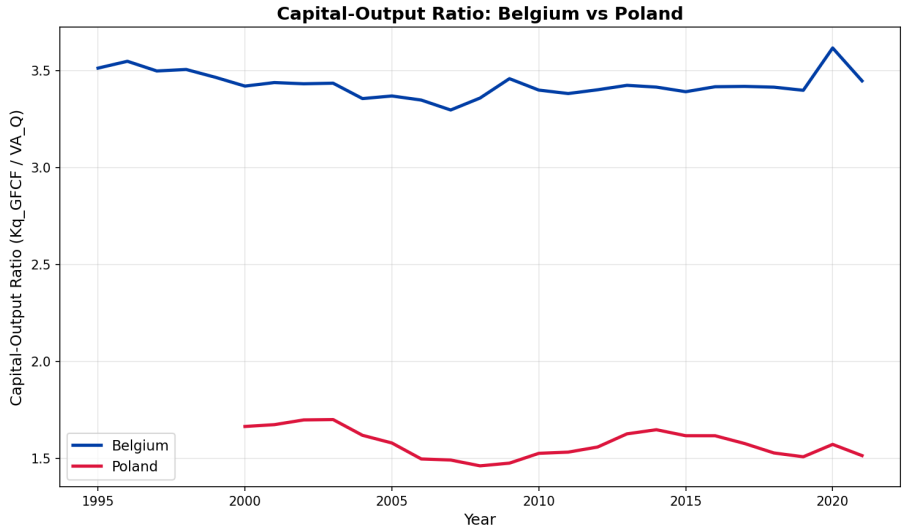


Figure 2: plots capital output ratio for belgium and Poland

Using $\frac{K_q^{GFCF}}{VA_Q}$, the average capital-output ratios are:

$$\overline{K/Y}_{BE} = 3.428 \quad (1995\text{–}2021), \qquad \overline{K/Y}_{PL} = 1.576 \quad (2000\text{–}2021).$$

Table 2: Average Capital-to-Output (K/Y) Ratios for belgium and poland

Country	Average K/Y ratio	Period
Belgium	3.428	$1995 - 2021$
Poland	1.576	$2000 - 2021$

Figure 2 shows a clear and persistent gap: Belgium's K/Y is much higher than Poland's over the full sample. Combining this with part (b), where $\alpha_{PL} > \alpha_{BE}$, implies that other parameters must differ. From $\frac{K}{Y} = \frac{\alpha}{r+\delta}$, Poland's lower K/Y suggests a higher effective $r + \delta$ (or transition/convergence effects). This is economically plausible for a catching-up economy, so the two findings are consistent within the model.